Xpert Cold Start: Dossier-ARR-PRISM

What we focused on

We prioritize relationship context, task discovery, conservative closure and restraint in interventions. A false completion hides work; an intrusive reminder can make an assistant unwelcome. We preserved the existing Python architecture and added targeted fixes, rather than rebuilding it. Selective inference avoids paying a model to read routine chatter. The deliverable is a batch JSON program, with no frontend or database.

Architecture

Contact dossiers combine seed relationships, saved-name aliases, chat membership and speaker-attributed samples. Organisation routing separates home and work and guards against promoting a client's business into the user's organisations. Goals require actual message evidence and user endorsement, with at most ten predictions.

Anchor-Retrieve-Reconcile retains original turns and retrieves local and cross-chat trails for commitments, requests, deadlines and decisions. Dates anchor to message time. Expired deadlines never imply completion. Acknowledgements, future intent, partial progress and unrelated later replies cannot establish execution. Cancellation can drop a loop; later failure evidence can reopen it. We enumerate all history; the checker alone applies the hidden scoring cutoff.

PRISM uses an asymmetric loss gate for assist/remind/leave. Its probabilities and loss values are hand-set assumptions, not empirically calibrated posteriors. External actions require approval; private preparation can proceed without it. Exact-ID corrections change relationships, chat ownership and task ownership/status while preserving unrelated predictions. Closing a corrected task requires later evidence.

Evaluation and actual cost

On 7 November we downloaded the official archives directly. Schema, checker and example match after newline normalization. Dev contains 4,502 messages /138 chats. Goals and interventions are unlabelled there: its denominator is 56, whereas the example is /74.

Run on official dev	Score /56	Provider USD /1k messages	Seconds
Original Arena, deterministic	18.68	0	7.00
Hardened, deterministic	19.80	0	8.33
12 November corrections, deterministic	28.95	0	4.93
Original Arena, GPT-4.1 mini live	19.48	0.01751	92.07
Hardened GPT-4.1 mini, serial /credit-limited	19.56	0.00942	789.08
Hardened GPT-4o mini, serial /credit-limited	19.80	0.00150	229.88

Exact tested OpenRouter IDs: **openai/gpt-4.1-mini** and **openai/gpt-4o-mini**. The original live run returned 128,022 input /17,264 output tokens in 39 responses, provider cost $0.0788312. Hardened serial runs returned 98,795 /13,075 tokens in 34 responses ($0.042390), and 38,565 /1,602 in 43 responses ($0.00674595), respectively. Relationship and loop-discovery prompts dominated original token use; detailed stage counts are in EVALUATION.md and the aggregate JSON. These are provider-reported costs, not price-table estimates; failed requests may omit usage.

Both hardened models suffered HTTP 402s and fallbacks, then permanent insufficient credits. Shared-account overlap confounds latency; these runs cannot rank models or select a grading model. The example scores 74/74 offline and in tested live mode. Historical synthetic scores use a test-aware mock and an incompatible rubric.

Where it fails

Latest deterministic dev loops have 17.99% precision, 67.57% recall and 28.41% F1: 50 matches among 278 predictions against 74 labels. There are zero false DONE among matched loops, 19 missed completions among matched loops (41 including undiscovered ones), and six owner errors. The starting hardened result matched 32/74 with 408 predictions. The official README warns that its loop labels are incomplete; unmatched predictions are not all necessarily invalid tasks.

Unseeded circle accuracy is 75.61%. Sparse contacts, dual roles, family/work vocabulary collisions, implicit obligations and ambiguous unquoted completions remain weak. Dev goal/intervention error rates cannot be measured. We do not hardcode identities or duplicate predictions to exploit repeated answer sources. Hidden performance is unknown.

Engineering decisions and limits

Inference never reads answers. Validated output is written atomically; unresolved schema/reference errors return nonzero. The client caches requests, bounds concurrency, reserves estimated cost and handles temporary in-flight-credit holds separately from permanent credit failures. A Linux watchdog caps execution below 30 minutes. All 114 local tests pass; Python 3.10/3.12/3.14 and clean-container CI remain configured.

A sparse synthetic 20k-message/320-chat run takes 1.19 s with a mock. A real-endpoint attempt took 4.02 s but all requests were rejected: successful dense-task network runtime remains unverified. The final relevance-only slow-pass refinement passed regressions but needs funded live re-evaluation. We retain Dossier-ARR-PRISM rather than add costly orchestration. Private data and credentials remain ignored. **Not ready to submit**: packaging is validated, but accuracy and clean model evidence need work.